TANKSHIFT – REVA BILL ANALYSIS

Part 12 of the Master Pack – savings estimate for the REVA campus pilot

STATUS: ESTIMATED. Built from assumed campus figures, not from actual bills. Every number below is tagged Verified / Assumed / Estimated.

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12.1 Why This Page Exists

The judges will ask "how much does one campus actually save?" The honest answer today is an estimate. This page shows the estimate, shows exactly which inputs it rests on, and lists the five numbers from REVA's real BESCOM HT bill that replace the assumptions. When the estate office provides six months of bills, only the yellow rows change; the method stays.

Tags used: Verified = taken from the KERC Electricity Tariff-2026 order (checked 27 Oct 2026). Assumed = a reasonable input for a private university campus in Bengaluru, not yet checked against REVA. Estimated = calculated from the above.

12.2 Inputs

Input	Value used	Tag	Where the real number comes from
Tariff category	HT-2(c)(ii) – private educational institution	Verified	Top of the BESCOM HT bill
Energy charge	₹7.00 per unit	Verified	KERC order; bill line "Energy charges"
ToD adders, Dec–Jun	6–9 am +₹1 · 9 am–6 pm ±0 · 6–10 pm +₹1 · 10 pm–6 am −₹1	Verified	KERC ToD table for HT-2
ToD adders, Jul–Nov	Zero in all slots	Verified	KERC order
Demand charge	₹380 per kVA of billing demand per month	Assumed	KERC range ₹350–380; bill line "Fixed/demand charges"
Contract demand	2,500 kVA	Assumed	Bill header
Billing demand	2,300 kVA (max recorded or 90 % of contract)	Assumed	Bill line "Billing demand"
Campus consumption	7,00,000 units per month	Assumed	Bill line "Total units"
Campus ToD split today	Shown on the bill as units per slot	Assumed	Bill ToD section – the single most useful line
Pump sets in pilot	3 × 15 HP (≈ 11 kW each), borewell + transfer	Assumed	Pump house nameplates

Input	Value used	Tag	Where the real number comes from
Pump energy per day	1,200 units (low 800, high 1,600)	Estimated	From water use: ~1,100 m^3/day × 0.55 kWh/m^3 borewell + 0.11 kWh/m^3 transfer; confirm with a clamp meter for one week
Pump share of campus load	5 % (3–7 %)	Estimated	Pump units ÷ campus units
Pump units in peak slots today	35 % (low 25 %, high 45 %)	Assumed	Float-switch pumps follow water use, which peaks 6–9 am and 6–10 pm; confirm with the clamp-meter week
Pump units in night slot today	15 % (10–20 %)	Assumed	Same
With TankShift: peak / day / night	2 % / 25 % / 73 % (high case 2 / 18 / 80)	Estimated	Firmware logic, Part 9: emergency fills only in peak
kVA moved off the site peak	20 kVA (low 10, high 37)	Assumed	37 kVA is all three pumps; 20 assumes pumps coincide with the campus peak about half the time
Hardware, 3 pump sets	₹30,000 installed (₹10,000 per set)	Estimated	Parts list, Part 6

12.3 Estimated Campus Bill Today (for scale)

Line	Monthly	Tag
Energy: 7,00,000 units × ₹7.00	₹49.0 lakh	Estimated
Demand: 2,300 kVA × ₹380	₹8.7 lakh	Estimated
ToD net (Dec–Jun, campus-wide, depends on slot split)	± ₹1–3 lakh	Estimated
Taxes, FAC, other	not estimated	–
Of which pumping (5 %)	≈ ₹2.5 lakh	Estimated

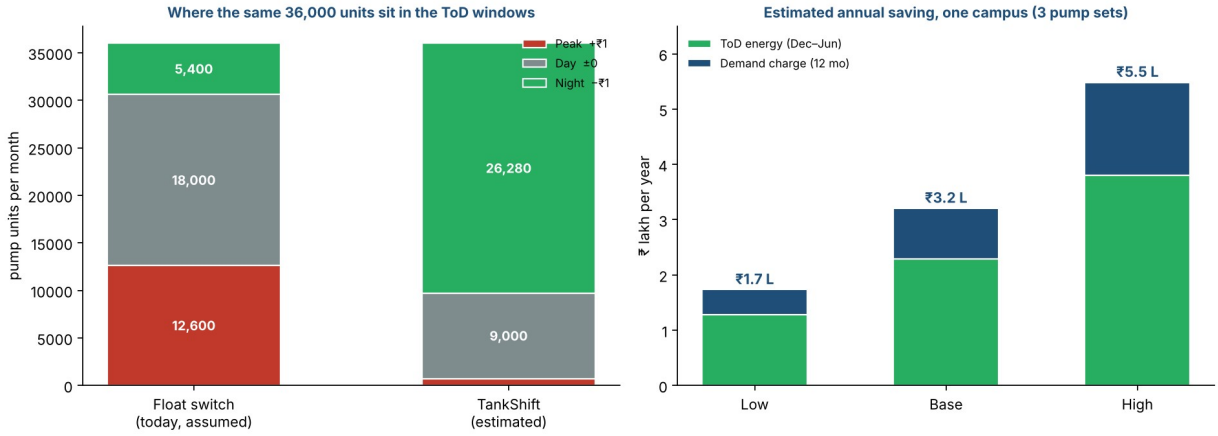
The point for the judges: pumping is a small slice of a large bill, but it is the one slice that can be moved by the clock without anyone noticing.

12.4 The Calculation

For each month in the ToD season, ToD cost = Σ (units in slot × adder). Saving = cost with float switches − cost with TankShift. Demand saving = kVA kept off the monthly peak × ₹380 × 12 months (applies all year, including monsoon).

	Low	Base	High	Tag
Pump units per month	24,000	36,000	48,000	Estimated
ToD cost today (per month)	+₹1,200	+₹7,200	+₹16,800	Estimated
ToD cost with TankShift (per month)	−₹17,040	−₹25,560	−₹37,440	Estimated
Monthly ToD saving	₹18,240	₹32,760	₹54,240	Estimated

	Low	Base	High	Tag
Effective swing per unit	₹0.76	₹0.91	₹1.13	Estimated
ToD saving per year (7 months)	₹1.28 lakh	₹2.29 lakh	₹3.80 lakh	Estimated
Demand saving per year (12 months)	₹0.46 lakh	₹0.91 lakh	₹1.69 lakh	Estimated
Total saving per year	₹1.73 lakh	₹3.21 lakh	₹5.48 lakh	Estimated
Hardware (3 sets)	₹30,000	₹30,000	₹30,000	Estimated
Payback	2.1 months	1.1 months	0.7 months	Estimated



ESTIMATED — built from assumed campus figures and the KERC 2025–28 ToD table; replace with REVA's actual HT bills before Demo Day.

12.5 What to Say on Stage

- Say: **"We estimate ₹2–5 lakh a year for a campus like ours, base case about ₹3 lakh, from a ₹30,000 install. The pilot replaces the estimate with the number on REVA's own bill."**
- Do not say ₹3–4 lakh energy plus ₹1–2 lakh demand as separate certainties; that was the earlier pack wording and it sits at the top of this range. Use the range and say "estimated".
- If asked what the estimate rests on: pump load of about 1,200 units a day (5 % of the campus), a third of it running in peak slots today, and the KERC ToD table. All three are on this page and all three get measured in week one of the pilot.

12.6 Five Numbers That Replace the Assumptions

Ask the estate office for the last six BESCOM HT bills (ideally Dec 2025 – May 2026, inside the ToD season). From each bill, copy:

1. Total units for the month.
2. Units in each ToD slot (the bill prints them separately for HT ToD consumers).
3. Billing demand (kVA) and contract demand (kVA).
4. Demand/fixed charge rate per kVA actually applied.
5. The ToD credit/debit line in rupees.

Then one week with a clamp-on energy meter on the three pump feeders gives pump units per day and their slot split. With those, rows tagged Assumed in 12.2 become Verified and the table in 12.4 prints the real number.

12.7 Changes to Make in the Other Documents

Document	Current wording	Change to
Master Pack, Slide 6 / Part 4.4 and 4.5	₹3–4 lakh/yr energy; ₹1–2 lakh/yr demand charges	"₹2–5 lakh/yr estimated (base ≈ ₹3 lakh); measured in the pilot"
Pitch Deck, slide 9 (Market)	One campus / year ₹3–5 lakh	Keep, it already shows a range; add "est." if space allows
Application Answers, impact and market	₹3–4 lakh energy and ₹1–2 lakh demand	"₹2–5 lakh a year estimated, base about ₹3 lakh"
Tabletop demo code, baselineRs	units × (BASE + 0.5)	Fine for a demo; it assumes 50 % of float-switch pumping in peak, the high case here

Sources

KERC Tariff Order 27 March 2025, Electricity Tariff-2026 annexure: HT-2(c)(ii) energy charge, ToD table for HT-2, demand charge range. Pump energy per m^3 from standard pump hydraulics ($\rho gh/\eta$, $\eta \approx 0.6$) at assumed heads of 100 m (borewell) and 20 m (transfer). All campus figures are assumptions pending the estate office bills.